Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
identify polynomials, monomials, binomials, and trinomials.			
determine the degree of polynomials.			
add and subtract monomials.			
add and subtract polynomials.			
evaluate a polynomial for a given value.			

ⓑ *If most of your checks were:*

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Whom can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.

10.2 Use Multiplication Properties of Exponents

Learning Objectives

By the end of this section, you will be able to:

- Simplify expressions with exponents
- Simplify expressions using the Product Property of Exponents
- Simplify expressions using the Power Property of Exponents
- Simplify expressions using the Product to a Power Property
- Simplify expressions by applying several properties
- Multiply monomials

☑ **BE PREPARED** 10.4 Before you get started, take this readiness quiz.

Simplify: $\frac{3}{4} \cdot \frac{3}{4}$.

If you missed the problem, review Example 4.25.

☑ **BE PREPARED** 10.5 Simplify: $(-2)(-2)(-2)$.

If you missed the problem, review Example 3.52.

Simplify Expressions with Exponents

Remember that an exponent indicates repeated multiplication of the same quantity. For example, 2^4 means to multiply four factors of 2, so 2^4 means $2 \cdot 2 \cdot 2 \cdot 2$. This format is known as **exponential notation**.

Exponential Notation

a^m ← exponent; a ← base

a^m means multiply m factors of a

$$a^m = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{m \text{ factors}}$$

This is read a to the mth power.

In the expression a^m, the exponent tells us how many times we use the base a as a factor.

7^3 $\quad$ $(-8)^5$

$7 \cdot 7 \cdot 7$ $\quad$ $(-8)(-8)(-8)(-8)(-8)$

3 factors $\quad$ 5 factors

Before we begin working with variable expressions containing exponents, let's simplify a few expressions involving only numbers.

EXAMPLE 10.11

Simplify:

ⓐ 5^3 ⓑ 9^1

Solution

ⓐ

	5^3
Multiply 3 factors of 5.	$5 \cdot 5 \cdot 5$
Simplify.	125

ⓑ

	9^1
Multiply 1 factor of 9.	9

TRY IT 10.21 Simplify:

ⓐ 4^3 ⓑ 11^1

TRY IT 10.22 Simplify:

ⓐ 3^4 ⓑ 21^1

EXAMPLE 10.12

Simplify:

ⓐ $\left(\frac{7}{8}\right)^2$ ⓑ $(0.74)^2$

Solution

ⓐ

	$\left(\frac{7}{8}\right)^2$

Multiply two factors.	$\left(\frac{7}{8}\right)\left(\frac{7}{8}\right)$
Simplify.	$\frac{49}{64}$

ⓑ

	$(0.74)^2$
Multiply two factors.	$(0.74)(0.74)$
Simplify.	0.5476

TRY IT 10.23 Simplify:

ⓐ $\left(\frac{5}{8}\right)^2$ ⓑ $(0.67)^2$

TRY IT 10.24 Simplify:

ⓐ $\left(\frac{2}{5}\right)^3$ ⓑ $(0.127)^2$

EXAMPLE 10.13

Simplify:

ⓐ $(-3)^4$ ⓑ -3^4

Solution

ⓐ

	$(-3)^4$
Multiply four factors of –3.	$(-3)(-3)(-3)(-3)$
Simplify.	81

ⓑ

	-3^4
Multiply two factors.	$-(3 \cdot 3 \cdot 3 \cdot 3)$
Simplify.	-81

Notice the similarities and differences in parts ⓐ and ⓑ. Why are the answers different? In part ⓐ the parentheses tell us to raise the (–3) to the 4th power. In part ⓑ we raise only the 3 to the 4th power and then find the opposite.

TRY IT 10.25 Simplify:

ⓐ $(-2)^4$ ⓑ -2^4

TRY IT 10.26 Simplify:

ⓐ $(-8)^2$ ⓑ -8^2

Simplify Expressions Using the Product Property of Exponents

You have seen that when you combine like terms by adding and subtracting, you need to have the same base with the same exponent. But when you multiply and divide, the exponents may be different, and sometimes the bases may be different, too. We'll derive the properties of exponents by looking for patterns in several examples. All the exponent properties hold true for any real numbers, but right now we will only use whole number exponents.

First, we will look at an example that leads to the Product Property.

	$x^2 \cdot x^3$
What does this mean? How many factors altogether?	$\underbrace{x \cdot x}_{2 \text{ factors}} \cdot \underbrace{x \cdot x \cdot x}_{3 \text{ factors}}$ $\underbrace{\qquad\qquad}_{5 \text{ factors}}$
So, we have	x^5
Notice that 5 is the sum of the exponents, 2 and 3.	$x^2 \cdot x^3$ is x^{2+3}, or x^5
We write:	$x^2 \cdot x^3$ x^{2+3} x^5

The base stayed the same and we added the exponents. This leads to the Product Property for Exponents.

Product Property of Exponents

If a is a real number and m, n are counting numbers, then

$$a^m \cdot a^n = a^{m+n}$$

To multiply with like bases, add the exponents.

An example with numbers helps to verify this property.

$$\begin{aligned} 2^2 \cdot 2^3 &\stackrel{?}{=} 2^{2+3} \\ 4 \cdot 8 &\stackrel{?}{=} 2^5 \\ 32 &= 32 \checkmark \end{aligned}$$

EXAMPLE 10.14

Simplify: $x^5 \cdot x^7$.

Solution

	$x^5 \cdot x^7$
Use the product property, $a^m \cdot a^n = a^{m+n}$.	x^{5+7}
Simplify.	x^{12}

TRY IT 10.27 Simplify: $x^7 \cdot x^8$.

TRY IT 10.28 Simplify: $x^5 \cdot x^{11}$.

EXAMPLE 10.15

Simplify: $b^4 \cdot b$.

Solution

	$b^4 \cdot b$
Rewrite, $b = b^1$.	$b^4 \cdot b^1$
Use the product property, $a^m \cdot a^n = a^{m+n}$.	b^{4+1}
Simplify.	b^5

TRY IT 10.29 Simplify: $p^9 \cdot p$.

TRY IT 10.30 Simplify: $m \cdot m^7$.

EXAMPLE 10.16

Simplify: $2^7 \cdot 2^9$.

Solution

	$2^7 \cdot 2^9$
Use the product property, $a^m \cdot a^n = a^{m+n}$.	2^{7+9}
Simplify.	2^{16}

TRY IT 10.31 Simplify: $6 \cdot 6^9$.

TRY IT 10.32 Simplify: $9^6 \cdot 9^9$.

EXAMPLE 10.17

Simplify: $y^{17} \cdot y^{23}$.

Solution

	$y^{17} \cdot y^{23}$
Notice, the bases are the same, so add the exponents.	y^{17+23}
Simplify.	y^{40}

TRY IT 10.33 Simplify: $y^{24} \cdot y^{19}$.

TRY IT 10.34 Simplify: $z^{15} \cdot z^{24}$.

We can extend the Product Property of Exponents to more than two factors.

EXAMPLE 10.18

Simplify: $x^3 \cdot x^4 \cdot x^2$.

Solution

	$x^3 \cdot x^4 \cdot x^2$
Add the exponents, since the bases are the same.	x^{3+4+2}
Simplify.	x^9

TRY IT 10.35 Simplify: $x^7 \cdot x^5 \cdot x^9$.

TRY IT 10.36 Simplify: $y^3 \cdot y^8 \cdot y^4$.

Simplify Expressions Using the Power Property of Exponents

Now let's look at an exponential expression that contains a power raised to a power. See if you can discover a general property.

	$(x^2)^3$
	$x^2 \cdot x^2 \cdot x^2$
What does this mean? How many factors altogether?	$\underbrace{x \cdot x}_{2 \text{ factors}} \cdot \underbrace{x \cdot x}_{2 \text{ factors}} \cdot \underbrace{x \cdot x}_{2 \text{ factors}}$ 6 factors
So, we have	x^6

Notice that 6 is the product of the exponents, 2 and 3.	$(x^2)^3$ is $x^{2\cdot 3}$ or x^6
We write:	$(x^2)^3$ $x^{2\cdot 3}$ x^6

We multiplied the exponents. This leads to the Power Property for Exponents.

Power Property of Exponents

If a is a real number and m, n are whole numbers, then

$$(a^m)^n = a^{m\cdot n}$$

To raise a power to a power, multiply the exponents.

An example with numbers helps to verify this property.

$$\begin{aligned}(5^2)^3 &\stackrel{?}{=} 5^{2\cdot 3}\\ (25)^3 &\stackrel{?}{=} 5^6\\ 15{,}625 &= 15{,}625\checkmark\end{aligned}$$

EXAMPLE 10.19

Simplify:

ⓐ $(x^5)^7$ ⓐ $(3^6)^8$

Solution

ⓐ

	$(x^5)^7$
Use the Power Property, $(a^m)^n = a^{m\cdot n}$.	$x^{5\cdot 7}$
Simplify.	x^{35}

ⓑ

	$(3^6)^8$
Use the Power Property, $(a^m)^n = a^{m\cdot n}$.	$3^{6\cdot 8}$
Simplify.	3^{48}

TRY IT 10.37 Simplify:

ⓐ $(x^7)^4$ ⓑ $(7^4)^8$

TRY IT 10.38 Simplify:

ⓐ $(x^6)^9$ ⓑ $(8^6)^7$

Simplify Expressions Using the Product to a Power Property

We will now look at an expression containing a product that is raised to a power. Look for a pattern.

	$(2x)^3$
What does this mean?	$2x \cdot 2x \cdot 2x$
We group the like factors together.	$2 \cdot 2 \cdot 2 \cdot x \cdot x \cdot x$
How many factors of 2 and of x?	$2^3 \cdot x^3$
Notice that each factor was raised to the power.	$(2x)^3$ is $2^3 \cdot x^3$
We write:	$(2x)^3$ $2^3 \cdot x^3$

The exponent applies to each of the factors. This leads to the Product to a Power Property for Exponents.

Product to a Power Property of Exponents

If a and b are real numbers and m is a whole number, then

$$(ab)^m = a^m b^m$$

To raise a product to a power, raise each factor to that power.

An example with numbers helps to verify this property:

$$\begin{aligned} (2 \cdot 3)^2 &\stackrel{?}{=} 2^2 \cdot 3^2 \\ 6^2 &\stackrel{?}{=} 4 \cdot 9 \\ 36 &= 36 \checkmark \end{aligned}$$

EXAMPLE 10.20

Simplify: $(-11x)^2$.

Solution

	$(-11x)^2$
Use the Power of a Product Property, $(ab)^m = a^m b^m$.	$(-11)^2 x^2$
Simplify.	$121x^2$

TRY IT 10.39 Simplify: $(-14x)^2$.

TRY IT 10.40 Simplify: $(-12a)^2$.

EXAMPLE 10.21

Simplify: $(3xy)^3$.

Solution

	$(3xy)^3$
Raise each factor to the third power.	$3^3x^3y^3$
Simplify.	$27x^3y^3$

TRY IT 10.41 Simplify: $(-4xy)^4$.

TRY IT 10.42 Simplify: $(6xy)^3$.

Simplify Expressions by Applying Several Properties

We now have three properties for multiplying expressions with exponents. Let's summarize them and then we'll do some examples that use more than one of the properties.

Properties of Exponents

If a, b are real numbers and m, n are whole numbers, then

Product Property	$a^m \cdot a^n = a^{m+n}$
Power Property	$(a^m)^n = a^{m \cdot n}$
Product to a Power Property	$(ab)^m = a^m b^m$

EXAMPLE 10.22

Simplify: $(x^2)^6(x^5)^4$.

Solution

	$(x^2)^6(x^5)^4$
Use the Power Property.	$x^{12} \cdot x^{20}$
Add the exponents.	x^{32}

TRY IT 10.43 Simplify: $(x^4)^3(x^7)^4$.

TRY IT 10.44 Simplify: $(y^9)^2(y^8)^3$.

EXAMPLE 10.23

Simplify: $(-7x^3y^4)^2$.

Solution

	$(-7x^3y^4)^2$
Take each factor to the second power.	$(-7)^2(x^3)^2(y^4)^2$
Use the Power Property.	$49x^6y^8$

TRY IT 10.45 Simplify: $(-8x^4y^7)^3$.

TRY IT 10.46 Simplify: $(-3a^5b^6)^4$.

EXAMPLE 10.24

Simplify: $(6n)^2(4n^3)$.

Solution

	$(6n)^2(4n^3)$
Raise $6n$ to the second power.	$6^2n^2 \cdot 4n^3$
Simplify.	$36n^2 \cdot 4n^3$
Use the Commutative Property.	$36 \cdot 4 \cdot n^2 \cdot n^3$
Multiply the constants and add the exponents.	$144n^5$

Notice that in the first monomial, the exponent was outside the parentheses and it applied to both factors inside. In the second monomial, the exponent was inside the parentheses and so it only applied to the *n*.

TRY IT 10.47 Simplify: $(7n)^2(2n^{12})$.

TRY IT 10.48 Simplify: $(4m)^2(3m^3)$.

EXAMPLE 10.25

Simplify: $(3p^2q)^4(2pq^2)^3$.

Solution

	$(3p^2q)^4(2pq^2)^3$
Use the Power of a Product Property.	$3^4(p^2)^4q^4 \cdot 2^3p^3(q^2)^3$

Use the Power Property.	$81p^8q^4 \cdot 8p^3q^6$
Use the Commutative Property.	$81 \cdot 8 \cdot p^8 \cdot p^3 \cdot q^4 \cdot q^6$
Multiply the constants and add the exponents for each variable.	$648p^{11}q^{10}$

TRY IT 10.49 Simplify: $(u^3v^2)^5(4uv^4)^3$.

TRY IT 10.50 Simplify: $(5x^2y^3)^2(3xy^4)^3$.

Multiply Monomials

Since a monomial is an algebraic expression, we can use the properties for simplifying expressions with exponents to multiply the monomials.

EXAMPLE 10.26

Multiply: $(4x^2)(-5x^3)$.

Solution

	$(4x^2)(-5x^3)$
Use the Commutative Property to rearrange the factors.	$4 \cdot (-5) \cdot x^2 \cdot x^3$
Multiply.	$-20x^5$

TRY IT 10.51 Multiply: $(7x^7)(-8x^4)$.

TRY IT 10.52 Multiply: $(-9y^4)(-6y^5)$.

EXAMPLE 10.27

Multiply: $\left(\frac{3}{4}c^3d\right)(12cd^2)$.

Solution

	$\left(\frac{3}{4}c^3d\right)(12cd^2)$
Use the Commutative Property to rearrange the factors.	$\frac{3}{4} \cdot 12 \cdot c^3 \cdot c \cdot d \cdot d^2$
Multiply.	$9c^4d^3$

TRY IT 10.53 Multiply: $\left(\frac{4}{5}m^4n^3\right)(15mn^3)$.

TRY IT 10.54 Multiply: $\left(\frac{2}{3}p^5q\right)(18p^6q^7)$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Exponent Properties (https://www.openstax.org/l/24expproperties)

Exponent Properties 2 (https://www.openstax.org/l/24expproperties2)

SECTION 10.2 EXERCISES

Practice Makes Perfect

Simplify Expressions with Exponents

In the following exercises, simplify each expression with exponents.

55. 4^5

56. 10^3

57. $\left(\frac{1}{2}\right)^2$

58. $\left(\frac{3}{5}\right)^2$

59. $(0.2)^3$

60. $(0.4)^3$

61. $(-5)^4$

62. $(-3)^5$

63. -5^4

64. -3^5

65. -10^4

66. -2^6

67. $\left(-\frac{2}{3}\right)^3$

68. $\left(-\frac{1}{4}\right)^4$

69. -0.5^2

70. -0.1^4

Simplify Expressions Using the Product Property of Exponents

In the following exercises, simplify each expression using the Product Property of Exponents.

71. $x^3 \cdot x^6$

72. $m^4 \cdot m^2$

73. $a \cdot a^4$

74. $y^{12} \cdot y$

75. $3^5 \cdot 3^9$

76. $5^{10} \cdot 5^6$

77. $z \cdot z^2 \cdot z^3$

78. $a \cdot a^3 \cdot a^5$

79. $x^a \cdot x^2$

80. $y^p \cdot y^3$

81. $y^a \cdot y^b$

82. $x^p \cdot x^q$

Simplify Expressions Using the Power Property of Exponents

In the following exercises, simplify each expression using the Power Property of Exponents.

83. $(u^4)^2$

84. $(x^2)^7$

85. $(y^5)^4$

86. $(a^3)^2$

87. $(10^2)^6$

88. $(2^8)^3$

89. $(x^{15})^6$

90. $(y^{12})^8$

91. $(x^2)^y$

92. $(y^3)^x$

93. $(5^x)^y$

94. $(7^a)^b$

Simplify Expressions Using the Product to a Power Property

In the following exercises, simplify each expression using the Product to a Power Property.

95. $(5a)^2$

96. $(7x)^2$

97. $(-6m)^3$

98. $(-9n)^3$

99. $(4rs)^2$

100. $(5ab)^3$

101. $(4xyz)^4$

102. $(-5abc)^3$

Simplify Expressions by Applying Several Properties

In the following exercises, simplify each expression.

103. $(x^2)^4 \cdot (x^3)^2$

104. $(y^4)^3 \cdot (y^5)^2$

105. $(a^2)^6 \cdot (a^3)^8$

106. $(b^7)^5 \cdot (b^2)^6$

107. $(3x)^2(5x)$

108. $(2y)^3(6y)$

109. $(5a)^2(2a)^3$

110. $(4b)^2(3b)^3$

111. $(2m^6)^3$

112. $(3y^2)^4$

113. $(10x^2y)^3$

114. $(2mn^4)^5$

115. $(-2a^3b^2)^4$

116. $(-10u^2v^4)^3$

117. $\left(\frac{2}{3}x^2y\right)^3$

118. $\left(\frac{7}{9}pq^4\right)^2$

119. $(8a^3)^2(2a)^4$

120. $(5r^2)^3(3r)^2$

121. $(10p^4)^3(5p^6)^2$

122. $(4x^3)^3(2x^5)^4$

123. $\left(\frac{1}{2}x^2y^3\right)^4(4x^5y^3)^2$

124. $\left(\frac{1}{3}m^3n^2\right)^4(9m^8n^3)^2$

125. $(3m^2n)^2(2mn^5)^4$

126. $(2pq^4)^3(5p^6q)^2$

Multiply Monomials

In the following exercises, multiply the following monomials.

127. $(12x^2)(-5x^4)$

128. $(-10y^3)(7y^2)$

129. $(-8u^6)(-9u)$

130. $(-6c^4)(-12c)$

131. $\left(\frac{1}{5}r^8\right)(20r^3)$

132. $\left(\frac{1}{4}a^5\right)(36a^2)$

133. $(4a^3b)(9a^2b^6)$

134. $(6m^4n^3)(7mn^5)$

135. $\left(\frac{4}{7}xy^2\right)(14xy^3)$

136. $\left(\frac{5}{8}u^3v\right)(24u^5v)$

137. $\left(\frac{2}{3}x^2y\right)\left(\frac{3}{4}xy^2\right)$

138. $\left(\frac{3}{5}m^3n^2\right)\left(\frac{5}{9}m^2n^3\right)$

Everyday Math

139. **Email** Janet emails a joke to six of her friends and tells them to forward it to six of their friends, who forward it to six of their friends, and so on. The number of people who receive the email on the second round is 6^2, on the third round is 6^3, as shown in the table. How many people will receive the email on the eighth round? Simplify the expression to show the number of people who receive the email.

Round	Number of people
1	6
2	6^2
3	6^3
…	…
8	?

140. **Salary** Raul's boss gives him a 5% raise every year on his birthday. This means that each year, Raul's salary is 1.05 times his last year's salary. If his original salary was \$40,000, his salary after 1 year was $\$40{,}000\,(1.05)$, after 2 years was $\$40{,}000(1.05)^2$, after 3 years was $\$40{,}000(1.05)^3$, as shown in the table below. What will Raul's salary be after 10 years? Simplify the expression, to show Raul's salary in dollars.

Year	Salary
1	$\$40{,}000\,(1.05)$
2	$\$40{,}000(1.05)^2$
3	$\$40{,}000(1.05)^3$
…	…
10	?